

Storage Backup Tool MVP Documentation

1. Project Overview

Project Name

SafeVault (Temporary MVP Name)

Project Type

Cross-platform desktop backup application for Windows and Linux.

Objective

Develop a lightweight and reliable backup tool that runs in the background and allows users to:

- Select folders or drives to back up
- Choose a destination storage location
- Schedule automatic backups
- Restore previous backups
- Run automatic incremental backups
- Compress and optionally encrypt backup data

The MVP focuses on reliability, simplicity, and automation.

2. MVP Goals

The MVP should:

- Work on both Windows and Linux
- Run continuously in the background
- Support scheduled backups
- Support incremental backups
- Provide a restore system
- Maintain backup history
- Use minimal system resources

The MVP will NOT initially include:

- Cloud backup
 - NAS support
 - Deduplication
 - AI scheduling
 - System image backup
 - Multi-device management
 - Mobile application
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3. Core Features

3.1 Backup Job Management

Users can create backup jobs.

Each backup job includes:

- Job name
- Source folder(s)
- Destination folder
- Schedule type
- Compression option
- Encryption option

Example:

Field	Value
Job Name	Office Backup
Source	D:/folder name
Destination	E:/folder name
Schedule	Daily
Compression	Enabled
Encryption	Disabled

3.2 Scheduled Backup

Supported schedules:

- Manual
- Daily
- Weekly
- Monthly

Features:

- Automatic execution
 - Background scheduling
 - Run at startup option
-

3.3 Incremental Backup

The system should copy only:

- New files
- Modified files

Comparison methods:

- Modified timestamp
- SHA256 file hash

Benefits:

- Faster backup
 - Reduced storage usage
 - Lower disk activity
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3.4 Restore System

Users can:

- Browse backup history
- Restore selected files
- Restore folders
- Restore complete backup snapshots

Restore options:

- Restore to original location

- Restore to custom location
-

3.5 Background Service

Application runs continuously in the background.

Windows:

- System tray application
- Startup execution

Linux:

- Background daemon
- Tray support

Features:

- Silent operation
 - Automatic scheduling
 - Notifications
-

3.6 Compression

Optional backup compression.

Supported formats:

- ZIP
- TAR.GZ (Linux future support)

Compression modes:

- No compression
 - Fast compression
 - Maximum compression
-

3.7 Encryption

Optional AES-256 encryption.

Features:

- Password-protected backups
- Encrypted backup archives
- Secure storage

Library:

- PyCryptodome
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3.8 Logs and Notifications

The system should log:

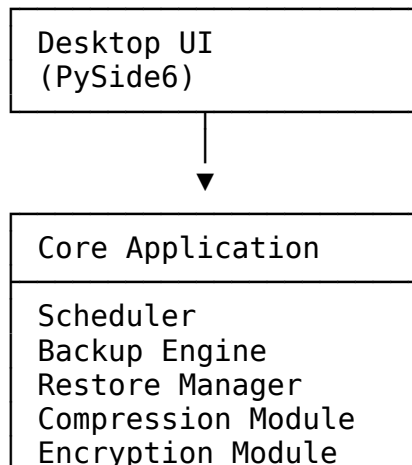
- Backup started
- Backup completed
- Backup failed
- File copy errors
- Restore operations

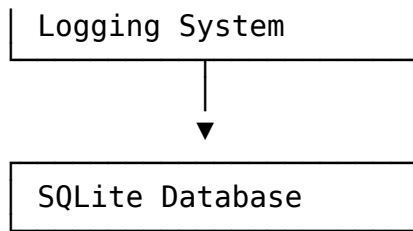
Notification types:

- Desktop notifications
 - System tray notifications
-

4. System Architecture

4.1 High-Level Architecture





5. Technology Stack

Component	Technology
Programming Language	Python
Desktop UI	PySide6
Database	SQLite
Scheduler	APScheduler
File Monitoring	watchdog
Encryption	PyCryptodome
Packaging	PyInstaller
Compression	zipfile / tarfile

6. Folder Structure

```
backup-tool/
├── app/
│   ├── ui/
│   │   ├── dashboard/
│   │   ├── backup_jobs/
│   │   ├── restore/
│   │   └── settings/
│   ├── backup_engine/
│   ├── scheduler/
│   ├── restore_engine/
│   ├── compression/
│   ├── encryption/
│   ├── monitoring/
│   ├── database/
│   ├── logs/
│   └── utils/
```

```
|
|— backups/
|— database/
|— logs/
|— tests/
|— requirements.txt
|— main.py
```

7. Backup Workflow

7.1 Full Backup Workflow

1. User creates backup job
 2. Scheduler triggers backup
 3. System scans source folders
 4. Metadata generated
 5. Files copied to destination
 6. Compression applied (optional)
 7. Encryption applied (optional)
 8. Database updated
 9. Logs generated
 10. Notification sent
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7.2 Incremental Backup Workflow

1. Scan source folder
 2. Compare with previous metadata
 3. Detect changed files
 4. Copy modified/new files only
 5. Update metadata database
 6. Generate logs
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8. Backup Storage Structure

Recommended Structure

```
Backups/
|
|— OfficeBackup/
|   |— 2026-05-25_10-00/
|   |— 2026-05-26_10-00/
|   |— latest/
```

└─ metadata/
└─ PersonalBackup/

9. Database Design

9.1 backup_jobs Table

```
CREATE TABLE backup_jobs (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    name TEXT NOT NULL,  
    source_path TEXT NOT NULL,  
    destination_path TEXT NOT NULL,  
    schedule_type TEXT,  
    compression_enabled INTEGER,  
    encryption_enabled INTEGER,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

9.2 backup_history Table

```
CREATE TABLE backup_history (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    job_id INTEGER,  
    backup_time TIMESTAMP,  
    status TEXT,  
    files_copied INTEGER,  
    backup_size INTEGER,  
    duration INTEGER  
);
```

9.3 file_metadata Table

```
CREATE TABLE file_metadata (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    job_id INTEGER,  
    file_path TEXT,  
    file_hash TEXT,  
    modified_time TIMESTAMP,  
    backup_version TEXT  
);
```

10. UI Design

10.1 Dashboard

Features:

- Active backup jobs
 - Last backup status
 - Next scheduled backup
 - Backup statistics
 - Storage usage
-

10.2 Create Backup Job Page

Fields:

- Job name
 - Source selection
 - Destination selection
 - Schedule selection
 - Compression toggle
 - Encryption toggle
-

10.3 Backup History Page

Display:

- Backup timestamps
 - Backup status
 - File counts
 - Backup size
 - Error logs
-

10.4 Restore Page

Features:

- Browse snapshots

- Select files/folders
 - Restore options
 - Restore progress
-

10.5 Settings Page

Features:

- Theme selection
 - Startup behavior
 - Notification settings
 - Backup retention settings
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11. Scheduler Design

Scheduler Responsibilities

The scheduler should:

- Monitor all backup jobs
- Trigger jobs at scheduled times
- Run jobs in background threads
- Prevent duplicate execution
- Generate notifications

Library:

- APScheduler
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12. File Monitoring

Real-Time Monitoring (Future MVP Enhancement)

Watch selected folders for:

- File creation
- File modification

- File deletion

Library:

- watchdog
-

13. Security Design

Encryption

Algorithm:

- AES-256

Features:

- Password-based encryption
 - Secure backup storage
 - Optional encryption mode
-

14. Error Handling

The system should handle:

- Missing drives
- Permission errors
- Locked files
- Storage full errors
- Interrupted backup processes

Features:

- Retry logic
 - Error logging
 - Skip inaccessible files
-

15. Performance Optimization

Optimization Strategies

- Chunked file copying
 - Background threads
 - Incremental scanning
 - Metadata caching
 - Avoid loading large files into memory
-

16. Packaging and Distribution

Windows

Package Type:

- EXE installer

Tool:

- PyInstaller
-

Linux

Package Types:

- AppImage
 - DEB package
-

17. Development Roadmap

Week 1

Project setup:

- Python environment setup
- PySide6 UI setup
- SQLite integration

- Folder structure creation
-

Week 2

Backup engine:

- File scanning
 - File copying
 - Incremental backup logic
 - Metadata storage
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Week 3

Scheduler and background system:

- APScheduler integration
 - Background threads
 - Tray application
 - Notifications
-

Week 4

Restore system:

- Snapshot browsing
 - File restore
 - Backup history
 - Error handling
-

Week 5

Security and optimization:

- Compression
- Encryption
- Performance improvements
- Bug fixing

18. Future Features

Planned future features:

- Cloud backup
 - NAS backup
 - Deduplication
 - System image backup
 - Ransomware protection
 - Remote monitoring
 - Multi-device dashboard
 - AI scheduling
 - Mobile companion app
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19. MVP Success Criteria

The MVP is considered successful if:

- Backups run automatically
 - Incremental backup works correctly
 - Restore system functions properly
 - Background execution is stable
 - Application works on Windows and Linux
 - Backup operations are reliable
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20. Conclusion

This MVP focuses on building a reliable, lightweight, and user-friendly backup solution for desktop systems.

The primary focus areas are:

- Reliability
- Automation

- Ease of use
- Cross-platform support
- Efficient incremental backups

The architecture is designed to support future expansion including cloud synchronization, deduplication, and enterprise-level backup features.